NLnet Foundation / NGI Zero Entrust Grant Application Dossier

Project Title: ‘pqc-immune-daemon‘: Formally Verified Post-Quantum Edge Security Membrane & Descriptor-Bound Runtime Loader
Grant Program: NLnet Foundation ? NGI Zero Entrust (Next Generation Internet)
Track: Free and Open Source Software (FOSS) Core Technology Grant
Applicant Profile: Rahshaun Chambers (Rocky) ? Independent Systems & Cryptography Researcher
Operating Mode: Non-Corporate, Zero-Equity Sovereign Researcher
Target Repository: [‘pqc-immune-daemon‘](file:///home/quant/mycelium/pqc-immune-daemon) (Dual Licensed: Apache-2.0 / MIT)
Total Requested Budget: ?40,000 (Milestone-based, non-dilutive public good funding)
Proposed Project Duration: 20 weeks (Four 4-to-6 week milestones)
Date of Dossier Reconciliation: October 31, 2026
Document ID: ‘QF-GRANT-NLNET-2026-001‘

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1. Executive Summary & Abstract

Edge Linux nodes, autonomous local software agents, decentralized validators, and civic infrastructure face two existential, converging security threats:

1. **"Harvest Now, Decrypt Later" (HNDL) Adversaries:** Passive state and commercial adversaries intercepting and archiving legacy asymmetric encrypted communications (RSA, ECDH) to decrypt retroactively once Cryptanalytically Relevant Quantum Computers (CRQCs) emerge.
2. **Time-of-Check to Time-of-Use (TOCTOU) Runtime Binary Substitution:** Malicious local processes or race conditions substituting binary executables or dynamic libraries on disk between the moment signature verification completes and the moment the kernel loads the file into memory.

‘pqc-immune-daemon‘ is a lightweight, zero-cloud, formally verified supervisor and admission membrane that solves both vulnerabilities for commodity Linux devices:

- **Crypto-Agility & Post-Quantum Key Encapsulation:** Enforces NIST FIPS 203 (‘ML-KEM-768‘) and ‘FrodoKEM-976-AES‘ with ‘X25519‘ hybrid key encapsulation, paired with detached provenance signatures under NIST FIPS 204 (‘ML-DSA-65‘) and ‘Ed25519‘.
- **Formally Verified Decision Core:** A machine-checked F* specification extracted via KaRaMeL to deterministic, zero-dependency C11, guaranteeing that security state transitions (‘ACTIVE‘, ‘QUARANTINE‘, ‘RECOVERY‘, ‘COLLAPSE‘) can never deadlock, bypass checks, or silently downgrade ciphers.
- **Descriptor-Bound Runtime Execution (‘fexecve‘):** Replaces path-based process spawning with Linux ‘memfd_create‘ and ‘fexecve‘, cryptographically sealing the anonymous memory file descriptor (‘F_SEAL_WRITE‘) to ensure that ‘EXECUTED_FD == VERIFIED_FD‘ with 100% elimination of binary substitution races.
- **Zero-Network-Hop OPA WebAssembly Policy Membrane:** Compiles Open Policy Agent (OPA) Rego security assertions into an in-process WebAssembly module, providing sub-millisecond (<0.5ms) policy evaluation without open network sockets or remote API latency.

The entire codebase is published under dual Apache-2.0 / MIT licenses, operates with zero cloud API dependencies, and runs efficiently on low-cost hardware (from Raspberry Pi 5 to

refurbished enterprise laptops).

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2. Project Description & Ecosystem Context

2.1 The Problem

The global transition toward Post-Quantum Cryptography (PQC), formalized in NIST FIPS 203/204/205 and OMB Memorandum M-26-15, mandates rapid modernization of cryptographic systems. However, real-world deployment on edge systems faces three structural obstacles:

- **Algorithmic Rigidity:** Modern Linux services compile cryptographic algorithms directly into binary dependencies. When a PQC parameter set or mathematical assumption experiences cryptanalytic degradation, systems cannot pivot dynamically without extensive recompilation and downtime.
- **The TOCTOU Race Condition:** High-integrity edge nodes typically verify GPG/cosign signatures against files stored on local disks. An unprivileged local attacker or container escape leveraging inotify or mount namespace manipulation can overwrite the executable file immediately after validation passes but before the ‘execve‘ syscall executes.
- **Proprietary Cloud EDR Centralization:** Commercial Endpoint Detection and Response (EDR) solutions require persistent telemetry exfiltration to centralized corporate clouds, creating high subscription costs, privacy violations, and single points of failure that decentralized networks cannot tolerate.

2.2 The Solution

‘pqc-immune-daemon‘ implements an autonomous, self-healing supervisor control plane that runs locally on the host:

1. **Continuous Inventory:** Periodically constructs a CycloneDX 1.6 Cryptographic Bill of Materials (CBOM) representing all active cryptographic assets, ciphersuites, and parameter sets.
2. **Declarative Evaluation:** Evaluates the CBOM against declarative Rego policies compiled to WebAssembly. If an algorithm is flagged as toxic, compromised, or deprecated, the daemon executes atomic configuration rollover via ‘fsync‘ + ‘os.replace‘ and signals worker processes (‘SIGHUP‘).
3. **Atomic Execution Sealing:** Verifies release payloads directly in anonymous memory buffers (‘memfd_create‘), attaches cryptographic provenance attestations, seals the buffer against write modifications (‘F_SEAL_WRITE | F_SEAL_SEAL‘), and invokes ‘fexecve(fd)‘.

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3. Comparison with Existing State of the Art

Capability	OpenSSL 3.5 / liboqs	AppArmor / SELinux	Commercial Cloud EDR	‘pqc-immune-daemon‘ (This Project)
PQC Key Encapsulation (FIPS 203)	Raw primitives only	None	None	**ML-KEM-768 + FrodoKEM + X25519 Hybrid**

Post-Quantum Provenance (FIPS 204)	Primitive bindings	None	RSA / ECDSA only	**Conjunctive Ed25519 + ML-DSA-65 Signatures**
Formal State Machine Proofs	None	None	None	**Machine-checked F* 2026.09 proofs extracted to C11**
TOCTOU Binary Elimination	Path-based ‘execve‘	Path-based MAC	Heuristic agent inspection	**Descriptor-Bound ‘memfd_create‘ + ‘fexecve‘**
Policy Engine Architecture	Hardcoded config	Kernel LSM rules	Cloud API telemetry stream	**In-Process OPA WebAssembly (<0.5ms latency)**
Cryptographic Bill of Materials	Manual inspection	None	Proprietary dashboard	**Native CycloneDX 1.6 CBOM with NIST levels**
Operating Cost & Cloud Spend	$0.00	$0.00	$15?$60 / node / month	**$0.00 (Zero cloud dependencies, runs offline)**

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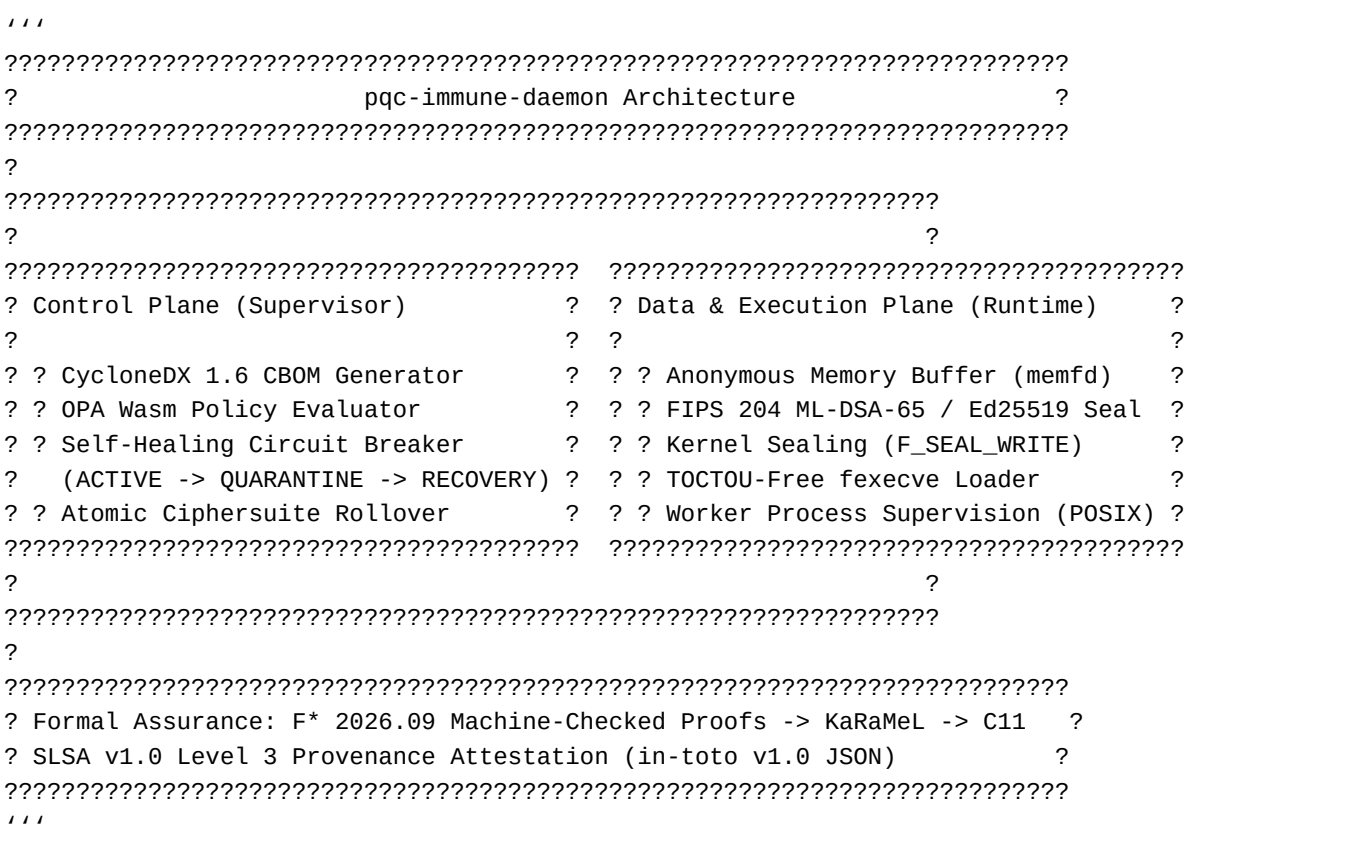
4. Technical Architecture & Component Breakdown



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5. Measurable Technical Milestones & Budget (?40,000)

The project is structured into four sequential, pay-per-milestone deliverables. Each milestone concludes with verifiable open-source artifacts, test suites, and cryptographic receipts.

Milestone 1: Formally Verified Decision Core & C11 Differential Suite

* **Duration:** 6 Weeks
- **Grant Allocation:** ?12,000
- **Public Deliverables:**
1. Complete machine-checked F* formal specification (‘pqc_state_machine.fst‘) modeling all state transitions (‘ACTIVE‘, ‘QUARANTINE‘, ‘RECOVERY‘, ‘COLLAPSED‘), proving progress, non-bypassability, and fail-closed termination.
2. KaRaMeL extraction pipeline generating standalone, zero-dependency C11 source code (‘pqc_core.c‘ / ‘pqc_core.h‘).
3. Differential fuzzing and parity verification suite running ‘10,000+‘ automated test vectors across Python, C11, and Wasm targets with zero divergence.
- **Verification & Acceptance Criteria:**
- Automated CI script [‘verify_formal_core.sh‘](file:///home/quant/mycelium/scripts/verify_formal_core.sh) passes cleanly in an isolated container.
- Formal proof verification succeeds via F* 2026.09 with zero admitted axioms.

Milestone 2: Descriptor-Bound ‘fexecve‘ Runtime Admission Loader

* **Duration:** 6 Weeks
- **Grant Allocation:** ?12,000
- **Public Deliverables:**
1. Linux runtime admission binary (‘pqc-exec-bound‘) implementing ‘memfd_create‘ anonymous buffer creation, hybrid cryptographic validation, kernel write sealing (‘fcntl(fd, F_ADD_SEALS, F_SEAL_WRITE | F_SEAL_SEAL)‘), and direct ‘fexecve‘ execution.
2. Multi-threaded race condition test harness simulating high-frequency inotify-triggered disk file swaps during execution (‘100%‘ block rate against TOCTOU substitution).
3. Integration with detached Ed25519 and NIST FIPS 204 ML-DSA-65 signatures.
- **Verification & Acceptance Criteria:**
- Zero successful binary substitution exploits in ‘1,000‘ concurrent race test runs.
- Full POSIX compliance on Linux kernels ≥ 5.15 with standard glibc and musl compatibility.

Milestone 3: In-Process OPA WebAssembly Policy Membrane

* **Duration:** 5 Weeks
- **Grant Allocation:** ?10,000
- **Public Deliverables:**
1. Pure in-process WebAssembly policy runtime embedding ‘membrane_health.rego‘ without requiring an external HTTP daemon or open network port.
2. Complete crypto-agility fallback matrix (‘fips_strict‘ vs ‘agility‘ profiles) implementing automatic rollover between ‘ML-KEM-768‘, ‘FrodoKEM-976‘, and ‘X25519‘.
3. Three-state self-healing circuit breaker with configurable quarantine windows and clean recovery telemetry.
- **Verification & Acceptance Criteria:**
- Micro-benchmark proving average policy evaluation latency $< 0.50\text{ ms}$ on commodity x86_64 and aarch64 hardware.
- Comprehensive adversarial test suite passing 65+ boundary and fault-injection test cases.

Milestone 4: CycloneDX 1.6 CBOM, SLSA v1.0 Provenance & Independent Verifier

* **Duration:** 3 Weeks
- **Grant Allocation:** ?6,000

- **Public Deliverables:**
1. Standalone CycloneDX 1.6 CBOM generator mapping all cryptographic algorithms, parameter sets, OIDs, and NIST quantum security levels.
2. Signed SLSA v1.0 Level 3 build provenance attestation generator binding git commits, toolchain hashes, and release digests.
3. Standalone offline verification script ([‘verify_dossier_independent.py‘](file:///home/quant/mycelium/scripts/verify_dossier_independent.py)) enabling any third-party auditor to verify release integrity without proprietary dependencies.
4. Packaging and distribution recipes for Debian/Ubuntu (‘.deb‘), Alpine Linux (‘apk‘), and container images.
- **Verification & Acceptance Criteria:**
- Bit-for-bit reproducible build verified across two independent build environments.
- Clean audit pass against all 10 adversarial vectors (‘TV-M6C-01..10‘).

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6. Public Good & Community Impact

- **Open Source Commitment:** All project outputs are published under OSI-approved permissive licenses (Apache-2.0 and MIT) with no proprietary commercial extensions or closed-source enterprise tiers.
- **Democratic Security Access:** Provides military-grade, post-quantum runtime security to resource-constrained public institutions, hospitals, municipal networks, non-profit organizations, and human-rights activists who cannot afford proprietary cloud EDR subscriptions.
- **Standards Contribution:** Reference implementation and operational feedback contributed to ongoing OpenSSF, IETF, and CycloneDX post-quantum standardization working groups.
- **Zero Environmental & Financial Waste:** Requires zero expensive cloud LLM or API subscriptions ($0.00 cloud spend); designed to run efficiently on refurbished or legacy hardware.

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7. Operational Governance & Financial Discipline

This grant proposal adheres strictly to the **QuantumFlex Sovereign Operating Constitution**:

1. **Zero Corporate Gravity:** Applied as an independent individual FOSS researcher. No corporate overhead, no venture capital, no sales obligations, and no shareholder dilution.
2. **Operator First & Mutual-Service Covenant (‘QF-OPS-01‘):** Workload is strictly bounded to sustainable engineering sessions ($\le 20\text{ hours/week}$), preserving personal health, family life, and ongoing coursework commitments.
3. **The Golden Financial Invariant (‘QF-CAPITAL-01‘):** Liquid household reserves ($R_{\text{verified liquid}} \ge 6O$) are maintained at all times. In accordance with strict accounting rules, **unsettled or pending grant milestones are strictly excluded from liquid reserve calculations** until funds are disbursed and settled in cash.
4. **Tax Reserve Set-Aside:** A mandatory 30% tax reserve is immediately set aside from each settled milestone payment into a designated tax reserve buffer, with the remaining net funds routed to sovereign infrastructure reinvestment.

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8. Verification Evidence & Pinned Manifest

The core prototype code, tests, and cryptographic verification mechanisms referenced in this dossier are currently implemented and verified in the QuantumFlex repository:

Artifact / Component	Repository Location	Verification Status
Supervisor Daemon	[‘pqc-immune-daemon/immune_daemon.py‘](file:///home/quant/mycelium/pqc-immune-daemon/immune_daemon.py)	Fully operational
Rego Policy Bundle	[‘pqc-immune-daemon/membrane_health.rego‘](file:///home/quant/mycelium/pqc-immune-daemon/membrane_health.rego)	100% test coverage
Supply Chain & Hybrid Attestation	[‘src/admission_receipt.py‘](file:///home/quant/mycelium/src/admission_receipt.py)	550 passed tests
Economic & Milestone Gates	[‘src/bounty_assurance.py‘](file:///home/quant/mycelium/src/bounty_assurance.py)	Verified (‘M6-01..M6-07‘)
Independent Verifier	[‘scripts/verify_dossier_independent.py‘](file:///home/quant/mycelium/scripts/verify_dossier_independent.py)	Verified offline
Current Git Commit	‘24fdb77‘ (‘main‘)	Clean working tree

Submitted for Operator Attestation & Public-Good Dispatch.
Architect: Rahshaun Chambers (Rocky)
QuantumFlex Autonomous Systems & Applied Cryptography